

Ribbon Font Neural Style Transfer for OpenType-SVG Font

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Figure 1: Neural style transferred OpenType-SVG colored font generated from the Source Han Serif by the proposed system.

ABSTRACT

We use existing machine learning neural style transfer model, differential rasterizer, for colored font design. The input of the proposed system is an existing TrueType font and the output is an neural style transferred OpenType-SVG color font. Each character glyph contains a series of vector graphics path elements. A neural style transferred glyph consists of dozens of random Bézier curves initially distributed among the glyph space, gradually converging to glyph stokes of a given character. In contract, the process of designing a font manually requires specialized skills for designers. They use mouse cursor to adjust strokes and glyph kerning for each character. It require hours to achieve a specific style. A Chinese font contains thousands of characters. Neural style transfer techniques can be used to speed up the process for designing decorative fonts. In addition, we add different colors to the curves in a an neural style transfer character for visual appealing. In order to support colors in a font, we choose OpenType-SVG format, allowing displaying colors in software like Illustrator or Inkscape. We open-source our code at <https://github.com/su8691/ribbon>.

KEYWORDS

machine learning, neural style transfer, SVG, OpenType-SVG, font

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1 NEURAL STYLE TRANSFER

Scalable Vector Graphics (SVG) has a feature that pixel graphics cannot have, the graphics can be scaled up or down without distortion as they are enlarged or reduced. The application of computer fonts is everywhere in life. However, in the field of font design, it is usually more difficult to handle ideographic characters with complex structures than phonetic characters with fewer characters. Therefore, how to efficiently design Chinese fonts with machine learning becomes a pressing issue. In this paper, we take SVG combined with Chinese character design as the starting point. We use a differentiable rasterizer [Li et al. 2020] for automatic font creation. The SVG font generation systems proposed in this paper provides an systematic way for SVG font design, educing manually adjustment by font designers and lowing time cost.

The system developed in this paper aims at generating neural style transferred glyph SVG images by processing glyph raster images extracted from a given base font. Then, these SVG images are integrated into an OpenType-SVG font. As shown in Figure 2, the system is divided into three parts. The first part is to extract Joint Photographic Experts Group (JPG) or Portable Network Graphics (PNG) raster images of given font; the second part is to generate the corresponding glyph SVG images based on the input raster images; the last part is to unify the generated SVG images, being packaged into a set of Chinese character fonts.

The neural style transferred SVG images generated by the proposed system can be used for not only TrueType font but also for OpenType-SVG font. The OpenType-SVG font format is officially supported by Adobe. As shown in Figure 3, it supports multiple glyph colors. In the other hand, the TrueType font format only supports black glyph color. In addition, Paint.NET and Inkscape are free open source software that support OpenType-SVG fonts. Both of them can read the fonts generated by the proposed system in this paper smoothly. Figure 1 shows the ribbon font generated by the proposed system in this paper.

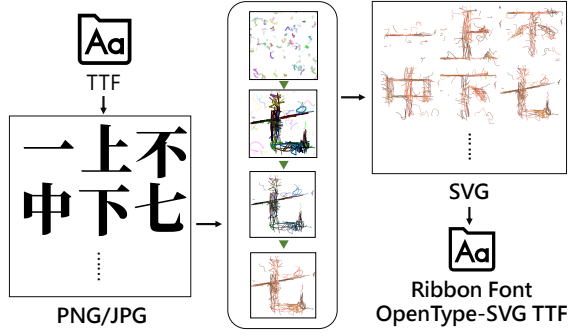


Figure 2: System architecture. Left: Raster images of all characters are extracted from a TrueType font. Center: Neural style transfer is performed to produce the corresponding SVG images for each character. Right: Add path color attribute in SVG images, combining them to create an OpenType-SVG font.

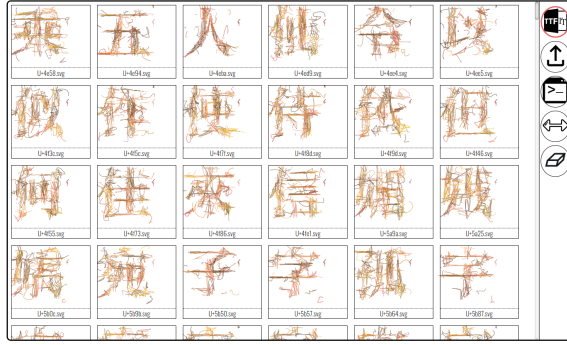


Figure 3: Screenshot of the OpenType-SVG font packaging system proposed in this paper, displaying the neural style transferred SVG images.

2 RESULTS AND CONCLUSIONS

In order to improve the aesthetic design of the font, we extracted the RGB values of 249 traditional Japanese colors to color the paths in a character. By controlling the color range and the transparency changes, different font color styles can be created as shown in Figure 4. In order to find out the optimal number of curves for generating appealing fonts, we experiment with different number of initial randomly distributed curves as shown in Figure 5 and Figure 6. As a result, we decide to set the number of the initial randomly distributed curves to be 100 curves and we set SVG paths stroke-width attribute to be 0.2.

In this study, we develop a OpenType-SVG font generation system using neural style transfer with differentiable rasterizer technique which connecting a raster graphics field to a vector graphics field. Case studies of open-source fonts, such as Source Han Serif and Source Han Sans, are used as the input base font for neural style transfer. Different font weights are used as the base fonts, including ExtraLight, Light, Regular, Medium, SemiBold, Bold, and Heavy. The output of the proposed system is the neural style transferred SVG images. These images can be combined together to create a



Figure 4: Comparison of SVG path coloring in a character with color range control and without color range control.

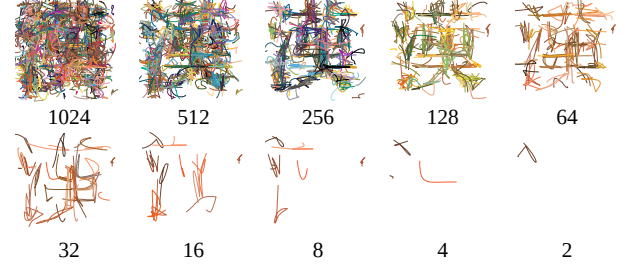


Figure 5: Comparison of the neural style transferred SVG images with different number of curves.



Figure 6: Comparison of the neural style transferred SVG images with different number of initial randomly distributed curves. Top row had 32 curves, middle row has 128 curves, and bottom row has 1024 curves.

set of Chinese fonts in OpenType-SVG font format. The new fonts produced by the proposed system in this paper can be used in Photoshop, Illustrator, and Inkscape.

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